

Explainer: *Thin Sets Are Not Equally Thin: Minimax Learning of Submanifold Integrals*

This note explains the paper’s main idea first informally and then in the paper’s formal language.

1 Feynman Approach for Intuition

Economists often care about parameters defined on sets that are tiny relative to the ambient data space: a point, a curve, a surface, or a level set. These sets have Lebesgue measure zero in the full space, so they are “thin.” But the paper’s central claim is that not all thin sets are equally hard statistically.

The correct measure of hardness is the intrinsic dimension of the target set. Learning an integral over a point is harder than over a curve, and learning over a curve is harder than over a surface, even if all of them are lower-dimensional than the ambient covariate space.

So the slogan “thin-set identification is irregular” is true but incomplete. The paper refines it: the degree of irregularity depends on the submanifold dimension m .

2 Formal Setup

The paper studies functionals of the form

$$\mathcal{G}(h_0) = \int_{\mathcal{M}} \phi(h_0(x), x) w(x) d\mathcal{H}^m(x),$$

where $\mathcal{M} \subset \mathbb{R}^d$ is an m -dimensional submanifold, $\mathcal{H}^m$ is the m -dimensional Hausdorff measure, and h_0 is an unknown function learned from data in the ambient d -dimensional space.

The linear benchmark is

$$L(h_0) = \int_{\mathcal{M}} h_0(x) w(x) d\mathcal{H}^m(x).$$

The paper also studies nonlinear submanifold integrals and upper contour functionals such as

$$V(h_0) = \int \mathbf{1}\{h_0(x) \geq 0\} w(x) dx.$$

The important asymptotic inputs are:

- ambient dimension d ;
- intrinsic submanifold dimension m ;
- smoothness s of h_0 .

3 Main Rigorous Result

If h_0 has Hölder smoothness s , then for regression and density settings the minimax rate for estimating these submanifold integral functionals is

$$n^{-\frac{s}{2s+d-m}}.$$

This is the paper’s headline result. The codimension $d - m$ is what determines how much statistical difficulty remains after integrating over the submanifold.

The logic is intuitive:

- the sample lives in d dimensions;
- the target lives on an m -dimensional object;
- what matters is how hard it is to recover information in the $d - m$ directions transverse to that object.

So when m increases, the problem becomes less irregular. In the extreme case $m = d$, the target is full-dimensional and the thin-set difficulty disappears.

4 Lower Bounds, Upper Bounds, and Why They Match

The paper is strong because it proves both sides.

Lower bounds. Section 3 proves that no estimator can beat the rate above uniformly over the relevant function classes. This is the minimax impossibility statement.

Upper bounds. Section 4 constructs sieve estimators that attain the same rate. So the rate is not just a lower bound; it is actually achievable.

That pairing is what makes the result complete. It tells us both what is impossible and what is optimal.

5 Inference

After establishing rate-optimal estimation, the paper develops inference for nonlinear submanifold integrals using sieve Riesz representers. The basic idea is to linearize the functional around h_0 and

then treat the leading linear term as the effective score. This yields asymptotic normality for t statistics and practical variance formulas.

This is especially valuable because many economically interesting functionals are nonlinear. The paper shows how to move from minimax rates to feasible inference rather than stopping at estimation theory.

6 Why the Paper Matters

The paper unifies a broad class of problems:

- integrals over known submanifolds;
- integrals over unknown level sets;
- value and welfare-type functionals whose derivatives reduce to boundary integrals;
- regression, density, and NPIV first stages.

The unifying message is that submanifold geometry enters first-order statistics through the intrinsic dimension m .

7 Bottom Line

The paper's Feynman-level message is: zero-measure targets are not all equally hard; their geometric dimension matters.

The rigorous version is: for submanifold integral functionals of an s -smooth object in ambient dimension d , the minimax rate is governed by $d - m$, and sieve methods can attain that rate while supporting asymptotically normal inference.